RESULTS

ICVGIP 2026 — Submission 485

Glacial Lake Segmentation using YOLOv11m-seg and SAM2 with Uncertainty-Guided Refinement

Annexure: Revised Tables and Figures

Table 1: Dataset Distribution

Dataset	Source	Images	Ground Truth
Training	Roboflow glacial-lake datasets (training split)	333	333
Validation	Roboflow glacial-lake datasets (validation split)	464	464
Test	Kaggle Glacial Lake Dataset (Sentinel-2, 10 m, Himalaya)	1640	1640
Total	All Dataset combined	2437	2437

Table 2: Performance of YOLOv11 variants and component-wise ablation

S.No.	Model / Configuration	IoU	Dice/F1	Prec.	Recall	Spec.	Pix. Acc.	Bal. Acc.	Params
1	YOLOv11n-seg	0.7511	0.8579	0.8950	0.8237	0.9877	0.9737	0.9067	2.9M
2	YOLOv11s-seg	0.7117	0.8315	0.8172	0.7605	0.8927	0.9703	0.8766	10.1M
3	YOLOv11m-seg	0.7021	0.8250	0.8368	0.7370	0.9947	0.9698	0.8659	22.4M
4	YOLOv11l-seg	0.7221	0.8386	0.8776	0.7794	0.9415	0.9711	0.8855	27.6M
5	YOLOv11x-seg	0.6978	0.8220	0.8627	0.7477	0.8424	0.9687	0.8700	62.1M
6	YOLOv11m-seg + SAM2	0.7600	0.8150	0.7520	0.8154	0.9868	0.9804	0.8991	103.2M
7	+ Boundary-aware box	0.6459	0.7849	0.7764	0.7936	0.9651	0.9423	0.8793	103.2M
8	+ Boundary-aware box and points	0.6804	0.8098	0.8024	0.8174	0.9692	0.9491	0.8933	103.2M
9	+ Random clicks (same number)	0.7092	0.8299	0.8499	0.8108	0.9781	0.9559	0.8944	103.2M
10	+ Distance-based clicks (same number)	0.7049	0.8269	0.8436	0.8109	0.9770	0.9550	0.8940	103.2M
11	**+ Uncertainty-guided correction (proposed)**	**0.7876**	**0.8816**	**0.7636**	**0.8126**	**0.9867**	**0.9802**	**0.8997**	**103.2M**
12	Full framework (box, points and correction)	0.7876	0.8816	0.7636	0.8126	0.9867	0.9802	0.8997	103.2M

Table 3: Comparison with other methods

Dataset	Method	IoU	Dice/F1	Precision	Recall
(a) Trained on the same data and tested on the same test set					
Kaggle test set (Sentinel-2)	U-Net	0.5145	0.6795	0.6058	0.7735
Kaggle test set (Sentinel-2)	DeepLabV3+	0.6403	0.7807	0.8706	0.7077
Kaggle test set (Sentinel-2)	YOLOv11m-seg + SAM2	0.7600	0.8150	0.7520	0.8154
Kaggle test set (Sentinel-2)	**Proposed framework**	**0.7876**	**0.8816**	**0.7636**	**0.8126**
(b) Reported in other papers on their own datasets (for reference only, not directly comparable)					

Dataset	Method	IoU	Dice/F1	Precision	Recall
Sentinel-1 & Sentinel-2	U-ViT	0.808	0.894	0.902	0.887
Sentinel-1 & Sentinel-2	U-Net	0.655	0.792	0.814	0.771
Sentinel-1 & Sentinel-2	DeepLabV3+	0.662	0.797	0.820	0.775
Sentinel-2	YOLOv5-seg	0.7160	0.7900	0.8245	0.7600
Sentinel-2	Adaptive SAM	0.7600	0.8073	0.8200	0.8113

Table 4: Performance after each iteration

Iteration	IoU	Dice/F1	Precision	Recall
Initial SAM2	0.7157	0.8343	0.8583	0.8116
1	0.7142	0.8333	0.8572	0.8107
2	0.7180	0.8359	0.8629	0.8104
3	0.7184	0.8361	0.8623	0.8115
4	0.7876	0.8816	0.7636	0.8126
5	0.7876	0.8816	0.7636	0.8126

Table 5: Validation of the uncertainty map against SAM2 error pixels

Uncertainty map	AUROC
Attention entropy (proposed)	**0.728**
Distance to boundary	0.693
Predictive entropy	0.559

Table 6: IoU and Dice under challenging conditions

Condition	Images	IoU (YOLOv11m-seg + SAM2)	Dice (YOLOv11m-seg + SAM2)	IoU (Proposed Framework)	Dice (Proposed Framework)
Small lakes	137	0.5774	0.7321	0.7024	0.8283
Medium lakes	136	0.7263	0.8415	0.7618	0.8686
Large lakes	137	0.7653	0.8670	0.8237	0.9062
Regular boundary	137	0.8869	0.9401	0.9126	0.9580
Irregular boundary	136	0.7396	0.8503	0.7864	0.8838
Highly irregular boundary	137	0.4837	0.6520	0.6218	0.7696
Single lake	60	0.9559	0.9775	0.9452	0.9753

Condition	Images	IoU (YOLOv11m-seg + SAM2)	Dice (YOLOv11m-seg + SAM2)	IoU (Proposed Framework)	Dice (Proposed Framework)
Multiple lakes	350	0.6794	0.8091	0.7425	0.8554
Snow, ice or cloud surroundings	280	0.7674	0.8684	0.8014	0.8928
Low-contrast boundary	137	0.6836	0.8121	0.6967	0.8251
Clear water	78	0.5859	0.7389	0.8691	0.9334

Table 7: Computational cost on an NVIDIA T4 GPU

Component	**Value**
YOLOv11m-seg	59.3 ms (123.3 GFLOPs)
SAM2 image encoder (once per image)	309.6 ms (646.8 GFLOPs)
SAM2 mask decoder (per iteration)	19.7 ms (3.66 GFLOPs)
Average corrective clicks per image	0.91
YOLOv11m-seg + SAM2 (total)	408.9 ms per image
Proposed framework (total)	**420.8 ms per image (2.38 images/s), 774.6 GFLOPs**
Peak GPU memory	1.37 GB
Parameters	103.2M (no additional parameters)

Table 8: Comparison with related prompt-refinement approaches (Page No. 8, Section 4.5)

Approach	**Needs training**	**Needs user clicks**	**Iterative correction**	**Uses model uncertainty**
RITM, SimpleClick (interactive segmentation)	Yes	Yes	Yes	No
FocSAM	Yes	Yes	Yes	No
VRP-SAM	Yes	No	No	No
Adaptive SAM	Yes	No	No	No
RS-SAM (remote-sensing SAM adaptation)	Yes	No	No	No
Detector-to-SAM pipeline (YOLO + SAM2)	No	No	No	No
Proposed framework	**No**	**No**	**Yes**	**Yes**

Table 9: Results for each Roboflow hard-case image

Case	Split	YOLOv11m-seg + SAM2 IoU	YOLOv11m-seg + SAM2 Dice	Proposed IoU	Proposed Dice	Proposed Precision	Proposed Recall	Clicks
Terrain shadow (Fig. 10a)	training	0.739	0.850	0.863	0.926	0.865	0.997	6
Terrain shadow (Fig. 10b)	test	0.780	0.876	0.800	0.889	0.840	0.943	10
Terrain shadow (Fig. 10c)	training	0.828	0.906	0.789	0.882	0.855	0.910	39
Snow and ice (Fig. 11a)	validation	0.503	0.669	0.515	0.680	0.523	0.971	10
Snow and ice (Fig. 11b)	test	0.936	0.967	0.926	0.962	0.978	0.946	0
Snow and ice (Fig. 11c)	test	0.974	0.987	0.964	0.982	0.996	0.967	3
Debris (Fig. 12a)	validation	0.716	0.835	0.718	0.836	0.895	0.784	14
Debris (Fig. 12b)	test	0.915	0.956	0.873	0.932	0.945	0.919	6
Debris (Fig. 12c)	validation	0.686	0.814	0.730	0.844	0.818	0.872	14
Mean (9 images)		**0.786**	**0.873**	**0.797**	**0.881**	**0.857**	**0.923**	**11.3**

Table 10: Comparison of primary and alternative data-split strategies and segmentation performance

Data-Split Strategy	Training	Validation	Test	IoU	Dice/F1	Precision	Recall
Primary split (used in study)	333 (13.7%)	464 (19.0%)	1640 (67.3%)	0.7876	0.8816	0.7636	0.8126
Alternative split (70:15:15)	1706 (70.0%)	366 (15.0%)	365 (15.0%)	0.7209	0.8378	0.8660	0.8114

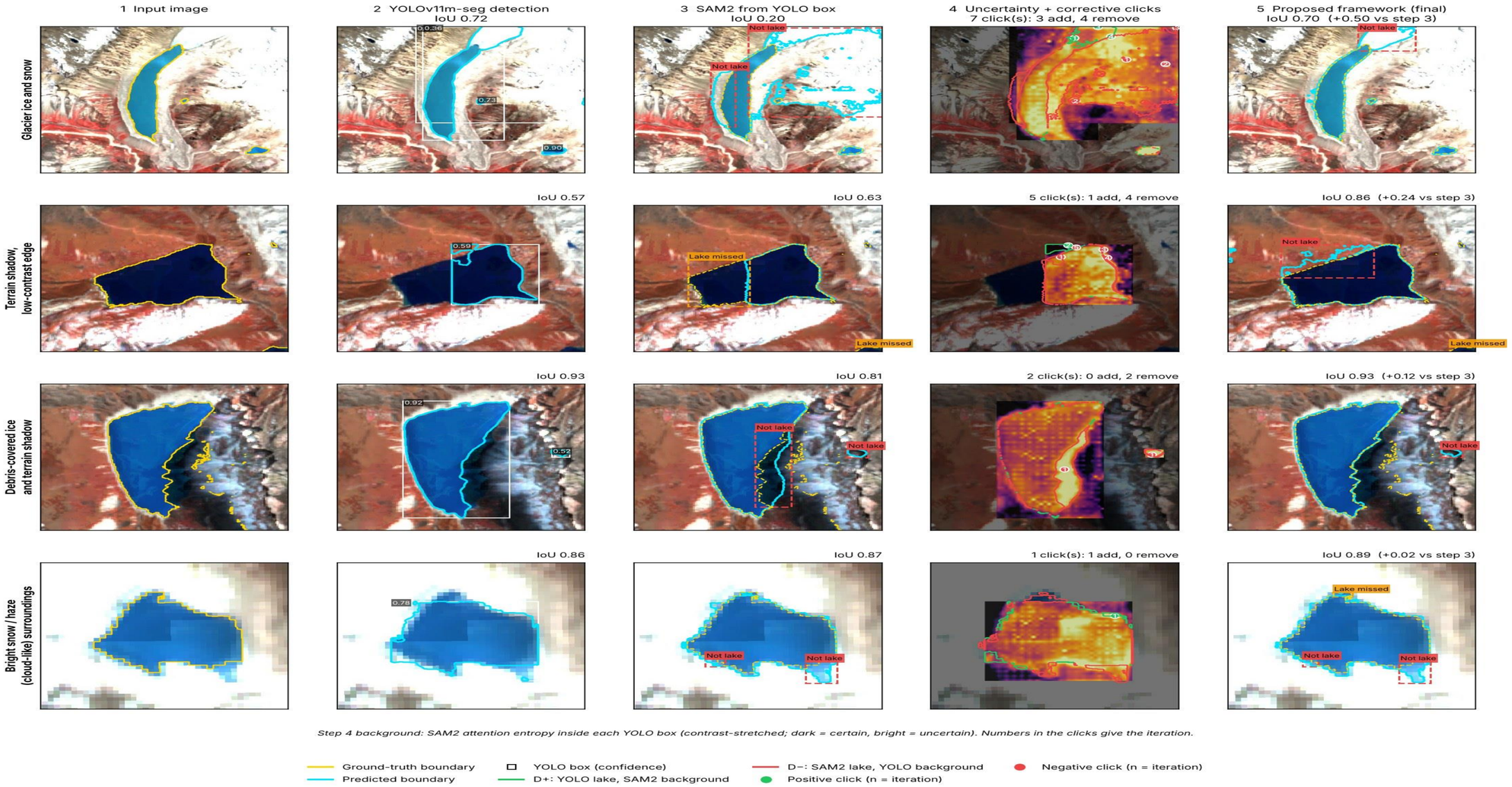


Fig. 1: Step-by-step correction under challenging conditions

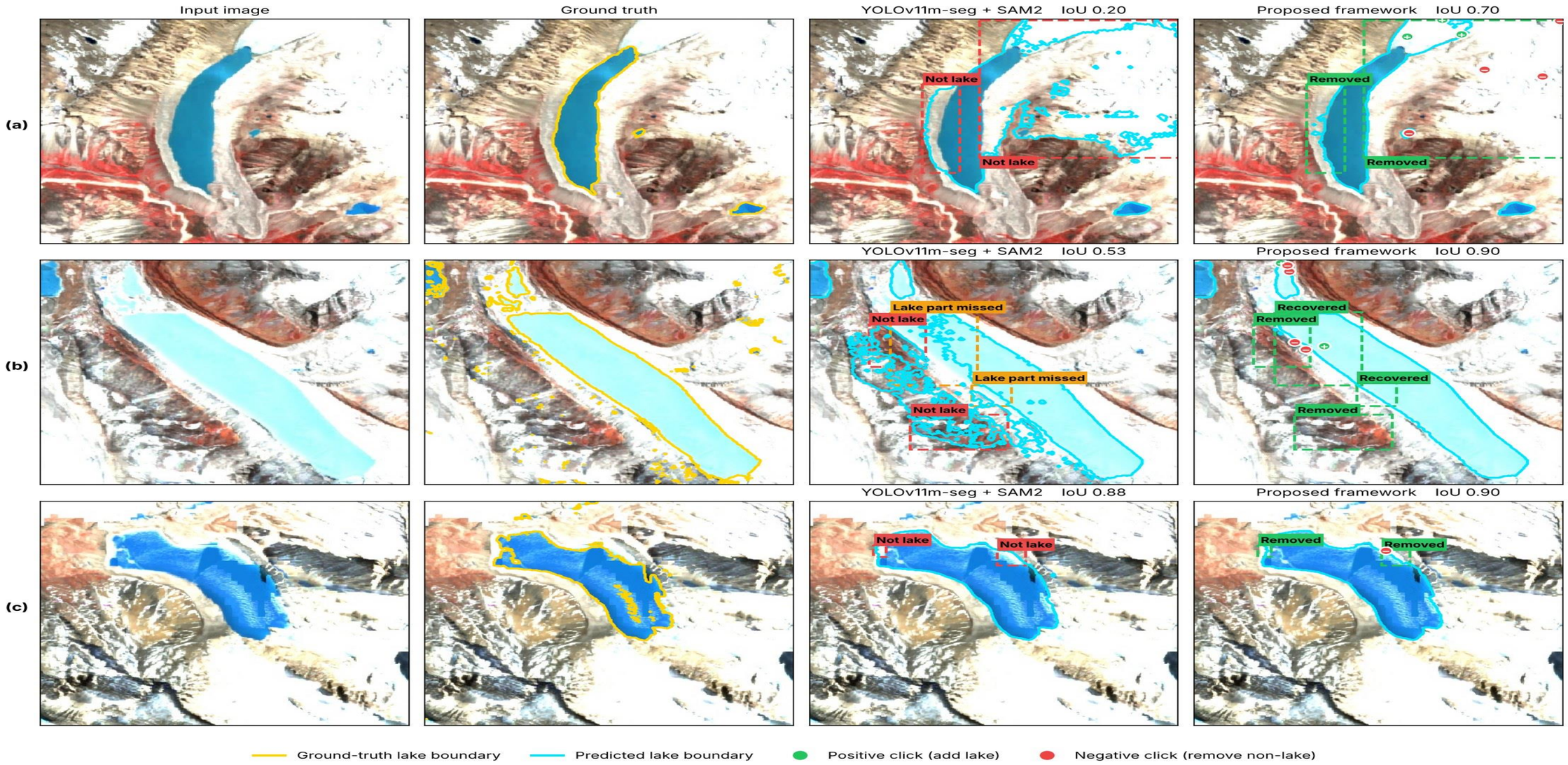


Fig. 2: Glacier ice and snow, Kaggle test images

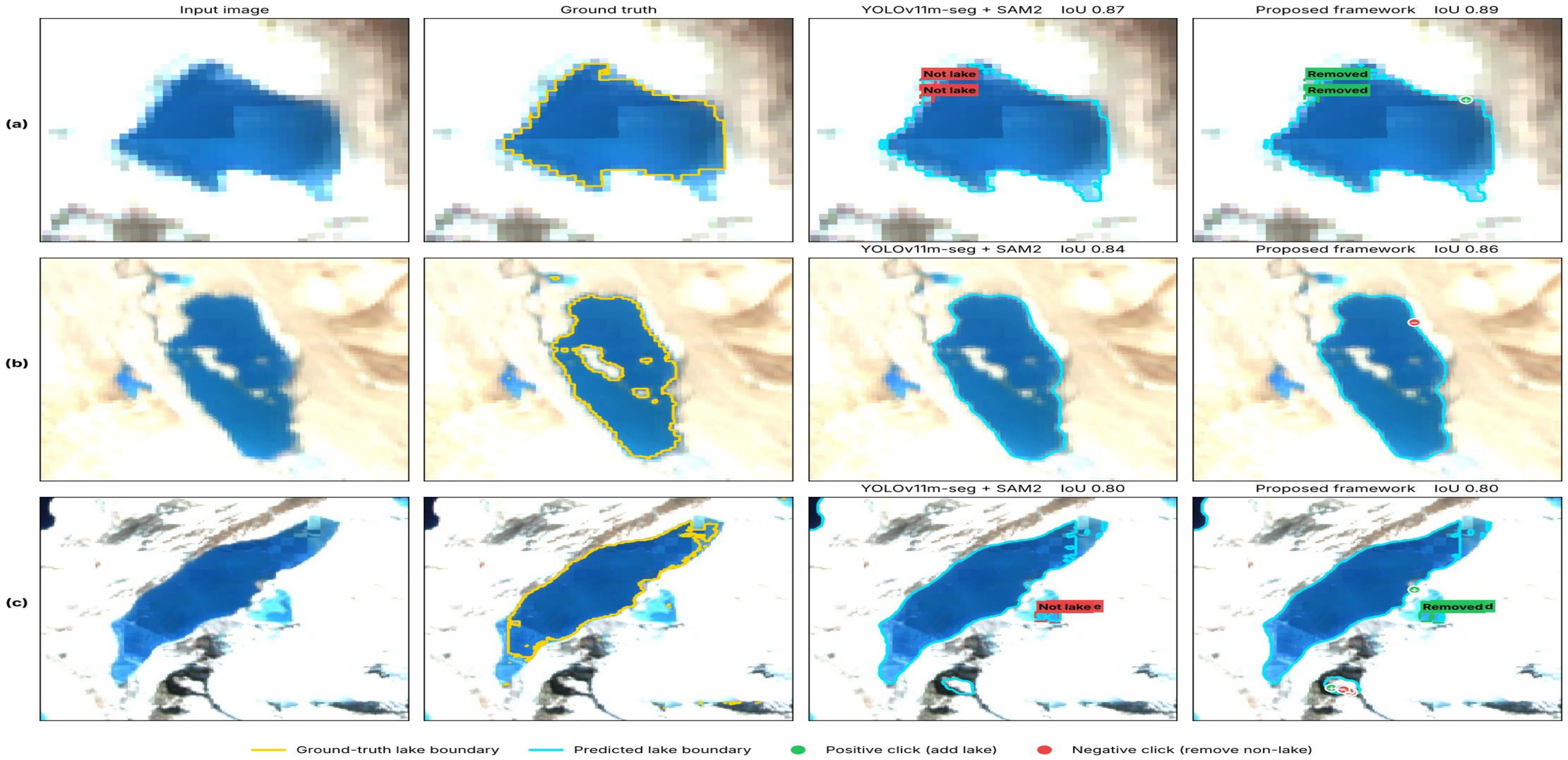


Fig. 3: Bright cloud- or haze-like surroundings, Kaggle test images

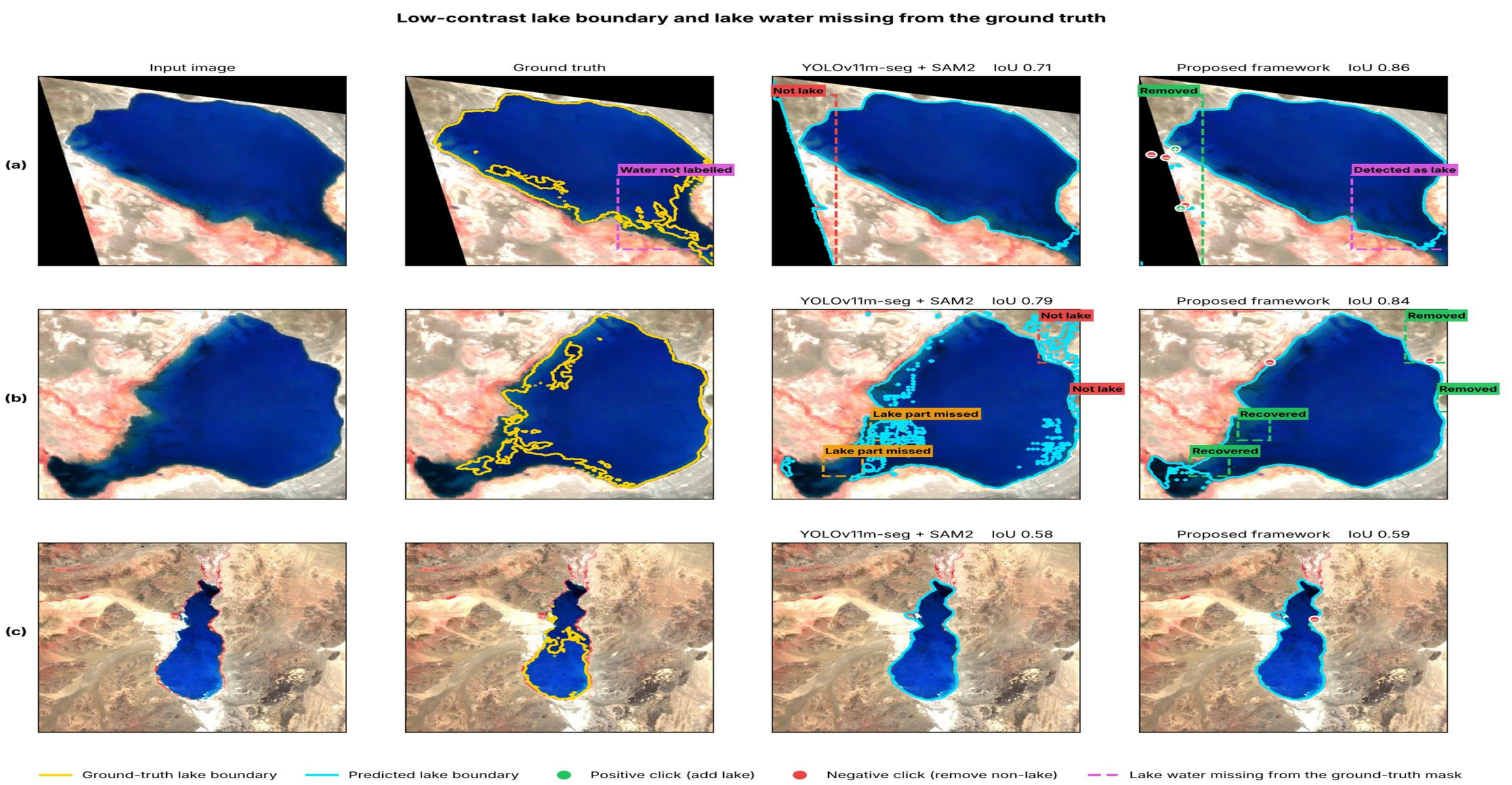


Fig. 4: Low-contrast lake edges and lake water missing from the ground truth, Kaggle test images

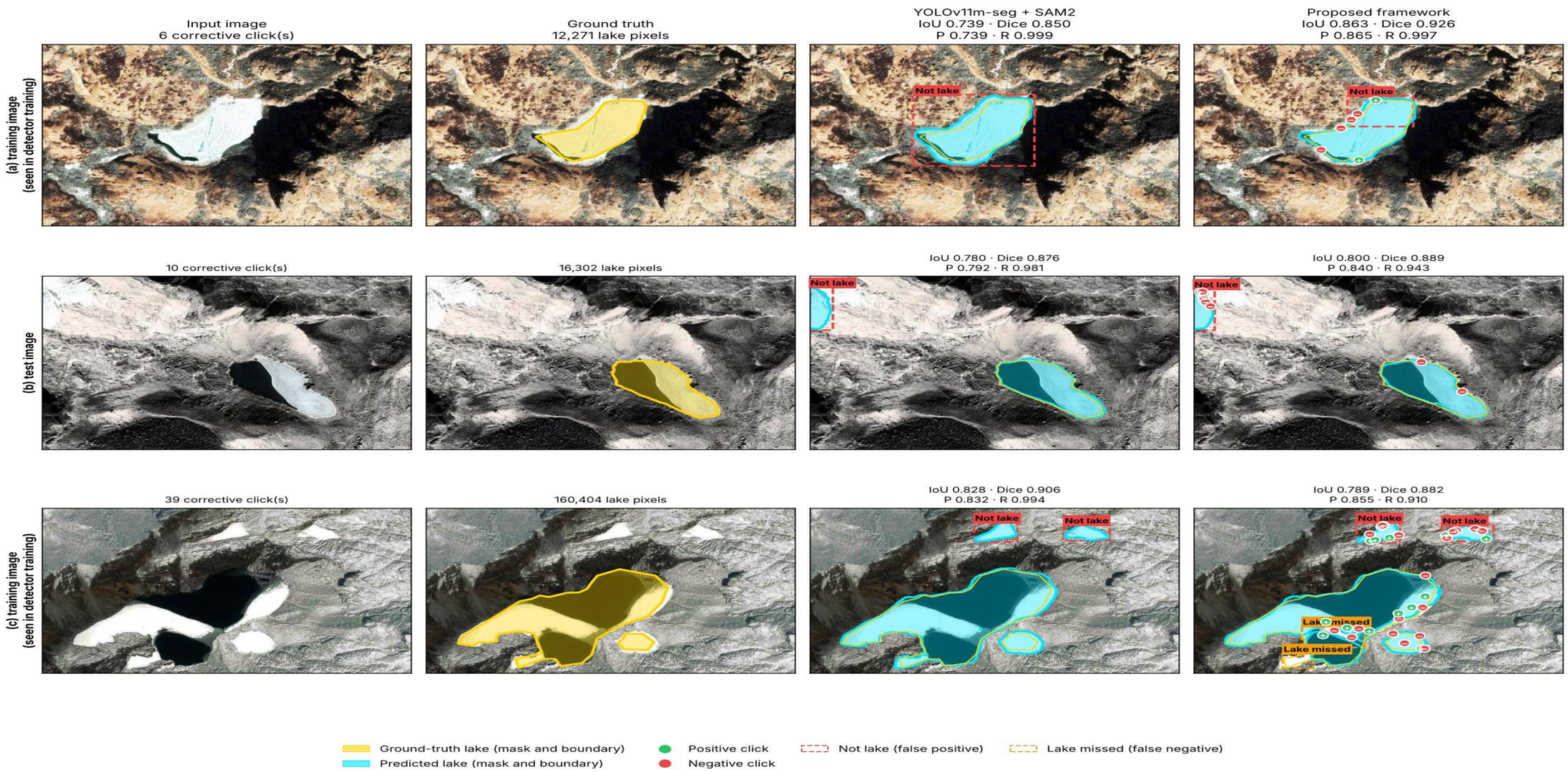


Fig. 5: Terrain shadow, Roboflow images

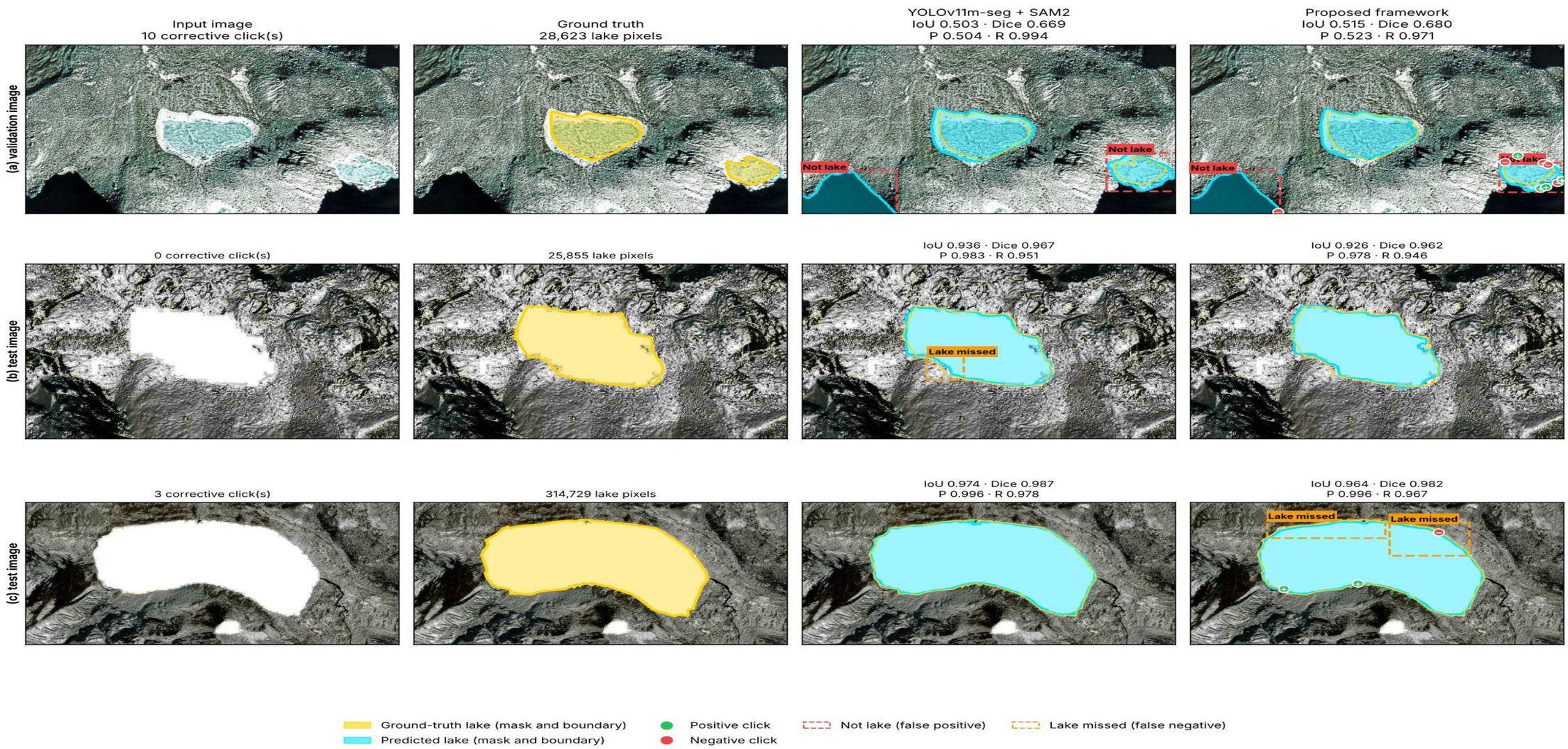


Fig. 6: Snow and ice cover, Roboflow images

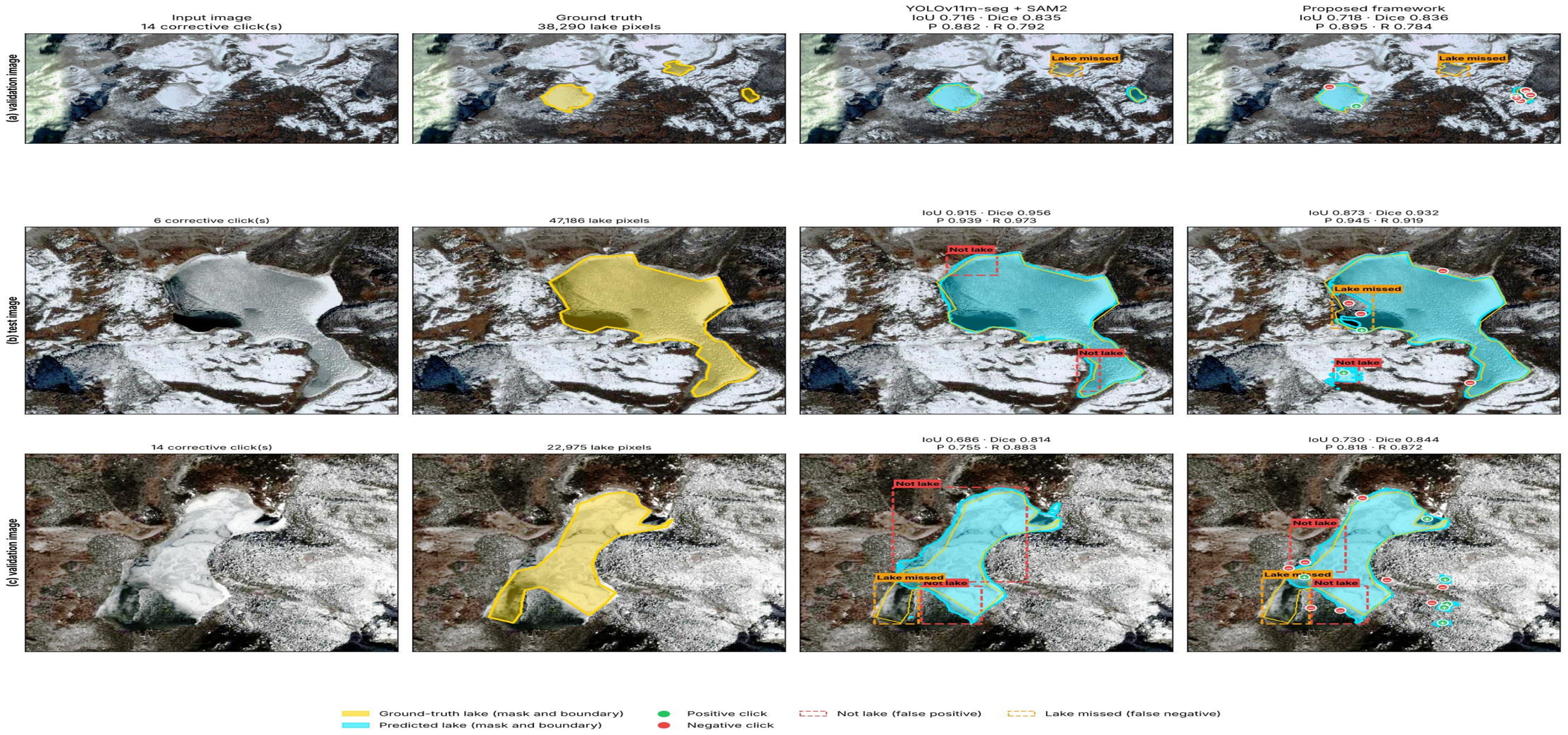


Fig. 7: Debris-covered surroundings, Roboflow images

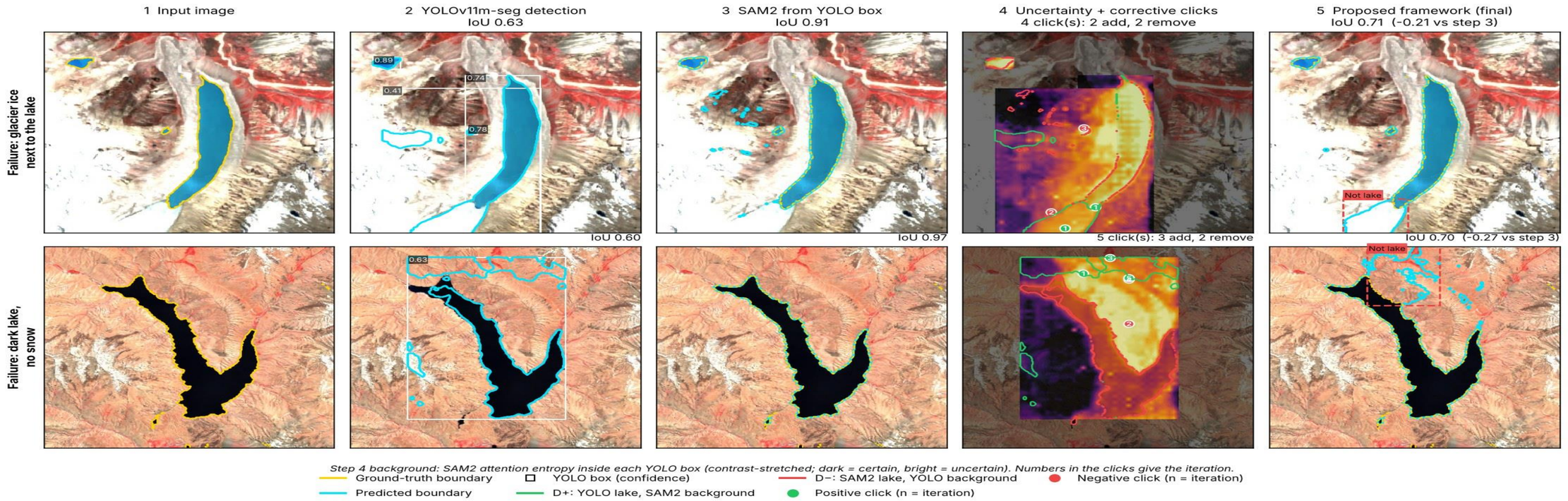


Fig. 8: Cases where the correction fails

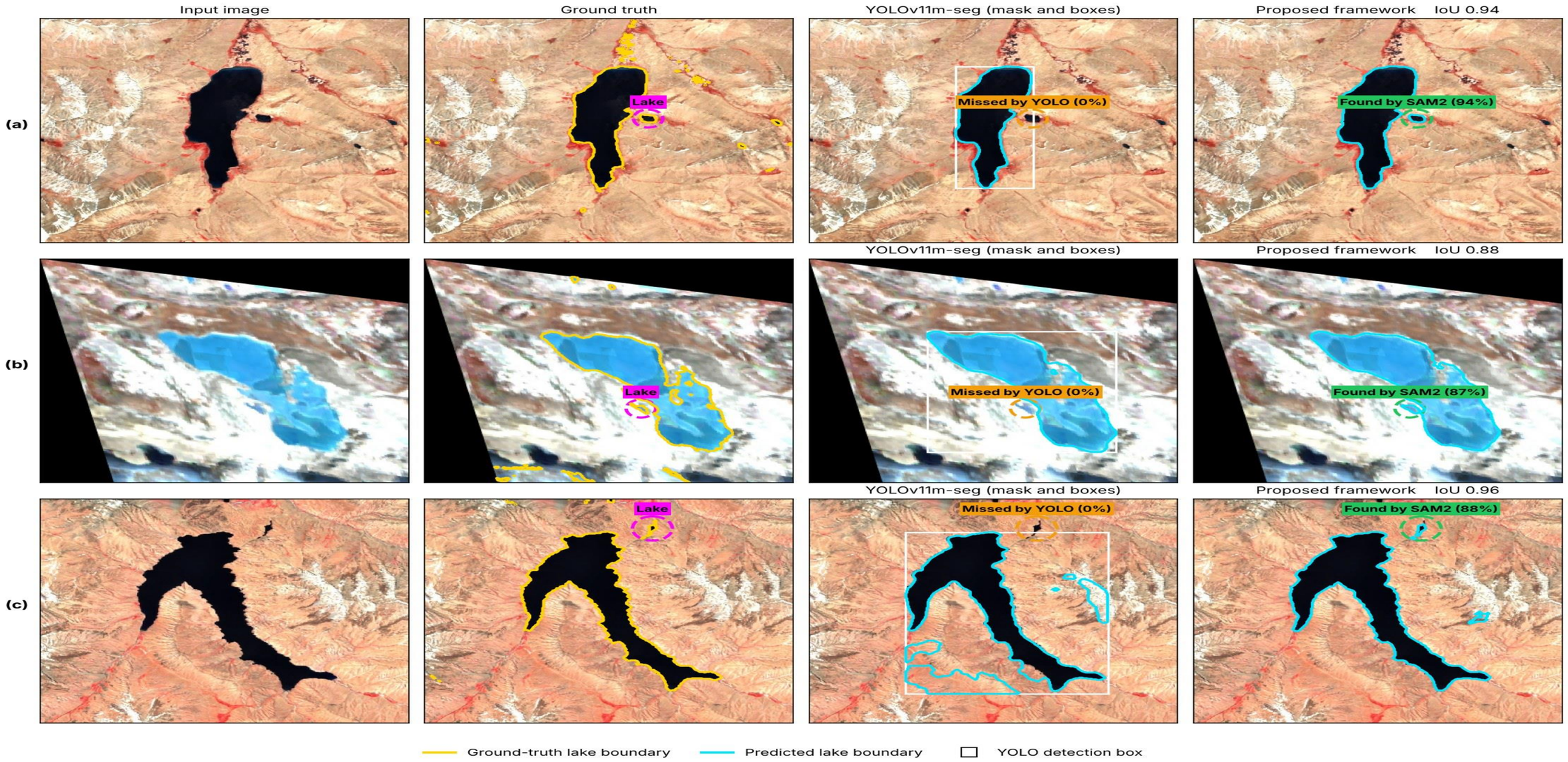


Fig. 9: Lakes missed by YOLOv11m-seg but found by SAM2